

Tuscany Pyroconvection + FWI – 3-Day Provincial Forecast – run 2026-07-20 00Z

Valid 2026-07-20 .. 2026-07-22. Hybrid: ICON-EU model levels + ICON-2I 2.2 km gate. Pyroconvection after Castellnou et al. (2022); Canadian FWI System (Van Wagner 1987).

Overview

Three valid days from the single **2026-07-20 00Z** run (day 0, +1, +2), using the day+1/+2 forecast steps in the same ICON-EU + ICON-2I datasets. The atmospheric pyroconvection classification is the hybrid product (ICON-EU native model levels for the profile – bulk-Richardson ABL, Bolton LCL, measured mixed-layer dtheta/dz, cap gamma-theta, ABL-top RH – with the 10 MW/m fuel gate on ICON-2I's 2.2 km surface fields over the Tuscany .lcp fuels). Two map series follow; per-province tables summarise both the pyroconvection potential and the Canadian fire-weather danger for each of the ten Tuscany provinces.

Fuel-gated pyroconvection (expected)

Potential pyroconvection (atmospheric upper bound)

Per-province pyroconvection potential metrics

Peak-of-day (09-18Z) statistics per province. *Pot peak* / *Gate peak* are the highest class reached (0 surface plume -> 4 deep pyroCu/pyroCb); *%pyroCu* is the maximum daytime areal coverage of class >= 2 (pyrocumulus), *%deep* of class 4. LCL/ABL, ABL depth and the minimum ABL-top RH are the noon column diagnostics over the province's classified land. Classes: convection_plume Convection plume, deep_pyrocu_pyrocb Deep pyroCu / pyroCb, overshooting_pyrocu Overshooting pyroCu, resilient_pyrocu Resilient pyroCu, surface_plume Surface plume.

Province	Day	Pot peak	Pot %pyroCu	Pot %deep	Gate peak	Gate %pyroCu	LCL/ABL	ABL m	RH-top min
Grosseto	2026-07-20	3	62.0	0.0	2	2.7	0.88	1635.0	29.0
Livorno	2026-07-20	4	48.0	1.5	2	1.8	1.01	1052.0	30.0
Siena	2026-07-20	2	93.5	0.0	2	6.8	0.83	2528.0	27.0
Pisa	2026-07-20	4	86.3	19.3	2	3.5	1.01	1681.0	24.0
Arezzo	2026-07-20	4	93.2	2.6	2	8.8	0.81	2661.0	23.0
Firenze	2026-07-20	4	91.6	10.7	4	6.1	0.83	2139.0	24.0
Prato	2026-07-20	4	98.9	24.2	4	9.9	0.8	2203.0	28.0
Pistoia	2026-07-20	4	91.6	18.1	2	8.4	0.77	2160.0	23.0
Lucca	2026-07-20	4	66.6	2.1	3	3.9	0.77	1990.0	23.0
Massa-Carrara	2026-07-20	4	48.8	6.2	3	2.2	0.74	2029.0	23.0
Grosseto	2026-07-21	4	77.9	0.2	2	2.8	0.94	1384.0	34.0
Livorno	2026-07-21	4	67.1	1.2	2	2.1	0.96	1035.0	38.0
Siena	2026-07-21	4	93.4	7.2	4	7.6	1.02	1882.0	38.0

Province	Day	Pot peak	Pot %pyroCu	Pot %deep	Gate peak	Gate %pyroCu	LCL/ABL	ABL m	RH-top min
Pisa	2026-07-21	3	98.4	0.0	2	6.2	0.96	1554.0	39.0
Arezzo	2026-07-21	4	96.5	74.6	4	13.3	0.94	1865.0	38.0
Firenze	2026-07-21	4	81.5	23.8	4	8.7	0.88	1405.0	43.0
Prato	2026-07-21	4	100.0	25.3	4	14.3	0.83	1349.0	55.0
Pistoia	2026-07-21	4	89.1	30.7	4	11.8	0.79	1527.0	51.0
Lucca	2026-07-21	4	71.0	5.5	4	6.0	0.84	1348.0	48.0
Massa-Carrara	2026-07-21	4	61.8	1.6	0	0.0	0.82	1187.0	57.0
Grosseto	2026-07-22	4	94.5	0.5	4	4.9	0.87	2617.0	27.0
Livorno	2026-07-22	4	86.7	2.7	3	3.6	0.97	2014.0	38.0
Siena	2026-07-22	4	100.0	0.6	3	3.5	0.83	2597.0	32.0
Pisa	2026-07-22	3	99.5	0.0	3	6.2	0.88	2504.0	34.0
Arezzo	2026-07-22	4	89.6	0.5	3	6.2	0.8	2211.0	42.0
Firenze	2026-07-22	4	95.0	0.4	3	5.0	0.82	2170.0	36.0
Prato	2026-07-22	3	100.0	0.0	3	6.6	0.81	2142.0	43.0
Pistoia	2026-07-22	4	100.0	8.8	4	5.5	0.81	2133.0	40.0
Lucca	2026-07-22	4	87.1	1.2	3	9.2	0.84	2009.0	27.0
Massa-Carrara	2026-07-22	4	73.0	0.6	3	4.3	0.77	2119.0	33.0

Per-province Canadian FWI System (12Z)

Province-mean of the full FWI System at solar noon on each valid day: the moisture **codes** FFMC (fine fuels), DMC (duff), DC (deep drought); the **indices** ISI (spread) and BUI (buildup); and the final **FWI**. The slow DMC/DC drought memory is seeded from local ERA5 reanalysis 2026-06-05..2026-07-02 (anchor DMC 98 / DC 261) + dry-spell persistence bridge 2026-07-03..2026-07-19 – **live CDS was unreachable at run time**, giving a run-day antecedent state of **FFMC 88.5 / DMC 144.9 / DC 416.0**; that state is then carried forward through the three forecast days on the run’s own 12Z weather. Forecast rain is taken as 0 (dry-spell assumption); a wetting event would lower FFMC/DMC and the indices. Danger classes follow the EFFIS FWI scale.

Province	Day	FFMC	DMC	DC	ISI	BUI	FWI	Danger
Grosseto	2026-07-20	89.7	148.9	425.7	8.8	158.9	34.0	high
Livorno	2026-07-20	88.6	148.0	425.4	6.8	158.3	28.6	high
Siena	2026-07-20	91.3	149.8	425.8	10.7	159.4	38.7	very high
Pisa	2026-07-20	89.8	149.0	425.6	9.1	158.9	35.0	high
Arezzo	2026-07-20	90.9	149.4	425.4	10.1	159.1	37.1	high
Firenze	2026-07-20	89.8	148.9	425.3	7.1	158.8	29.6	high
Prato	2026-07-20	89.7	148.8	425.3	6.4	158.8	27.4	high
Pistoia	2026-07-20	89.1	148.4	425.0	5.9	158.5	26.2	high
Lucca	2026-07-20	89.1	148.4	425.0	7.0	158.5	29.3	high
Massa-Carrara	2026-07-20	88.6	148.1	424.9	6.7	158.2	28.4	high
Grosseto	2026-07-21	89.9	152.8	435.3	10.3	162.8	37.9	high
Livorno	2026-07-21	88.8	151.4	434.8	6.8	161.9	28.8	high
Siena	2026-07-21	92.6	155.2	435.7	15.7	164.2	49.6	very high
Pisa	2026-07-21	91.1	153.7	435.2	12.1	163.2	42.1	very high
Arezzo	2026-07-21	92.5	154.8	435.1	14.1	163.8	46.5	very high

Province	Day	FFMC	DMC	DC	ISI	BUI	FWI	Danger
Firenze	2026-07-21	91.3	153.6	434.9	11.8	163.2	41.4	very high
Prato	2026-07-21	89.9	152.9	434.6	9.0	162.7	34.8	high
Pistoia	2026-07-21	89.5	152.2	434.2	9.1	162.2	34.9	high
Lucca	2026-07-21	89.5	152.0	434.1	8.4	162.1	33.1	high
Massa-Carrara	2026-07-21	88.5	151.1	433.7	6.9	161.5	28.9	high
Grosseto	2026-07-22	93.1	158.4	444.9	14.1	167.6	46.5	very high
Livorno	2026-07-22	92.4	156.7	444.4	12.1	166.5	42.2	very high
Siena	2026-07-22	93.1	160.4	445.0	12.0	168.7	42.2	very high
Pisa	2026-07-22	93.0	159.1	444.8	11.2	168.0	40.5	very high
Arezzo	2026-07-22	92.1	159.0	443.8	11.8	167.8	41.8	very high
Firenze	2026-07-22	91.5	158.0	443.8	11.5	167.2	40.7	very high
Prato	2026-07-22	90.9	157.1	443.4	10.8	166.6	39.0	very high
Pistoia	2026-07-22	90.3	156.1	442.8	9.7	166.0	36.2	high
Lucca	2026-07-22	90.6	156.2	442.8	9.0	166.0	34.6	high
Massa-Carrara	2026-07-22	89.6	154.8	442.2	7.6	165.1	30.8	high

Generated by scripts/forecast_3day_province_report.py. Pyroconvection: pyflam.pyroconvection_type (hybrid). FWI: pyflam.fwi + pyflam.fire_weather_history (ERA5/SIR spin-up). CSV tables accompany this report in the same folder.

Tuscany pyroconvection -- 3-day forecast -- FUEL-GATED -- expected (fire power ≥ 10 MW/m on the Tuscany fuels)
 ICON-EU model levels + ICON-2I 2.2 km gate -- run 2026-07-20 00Z

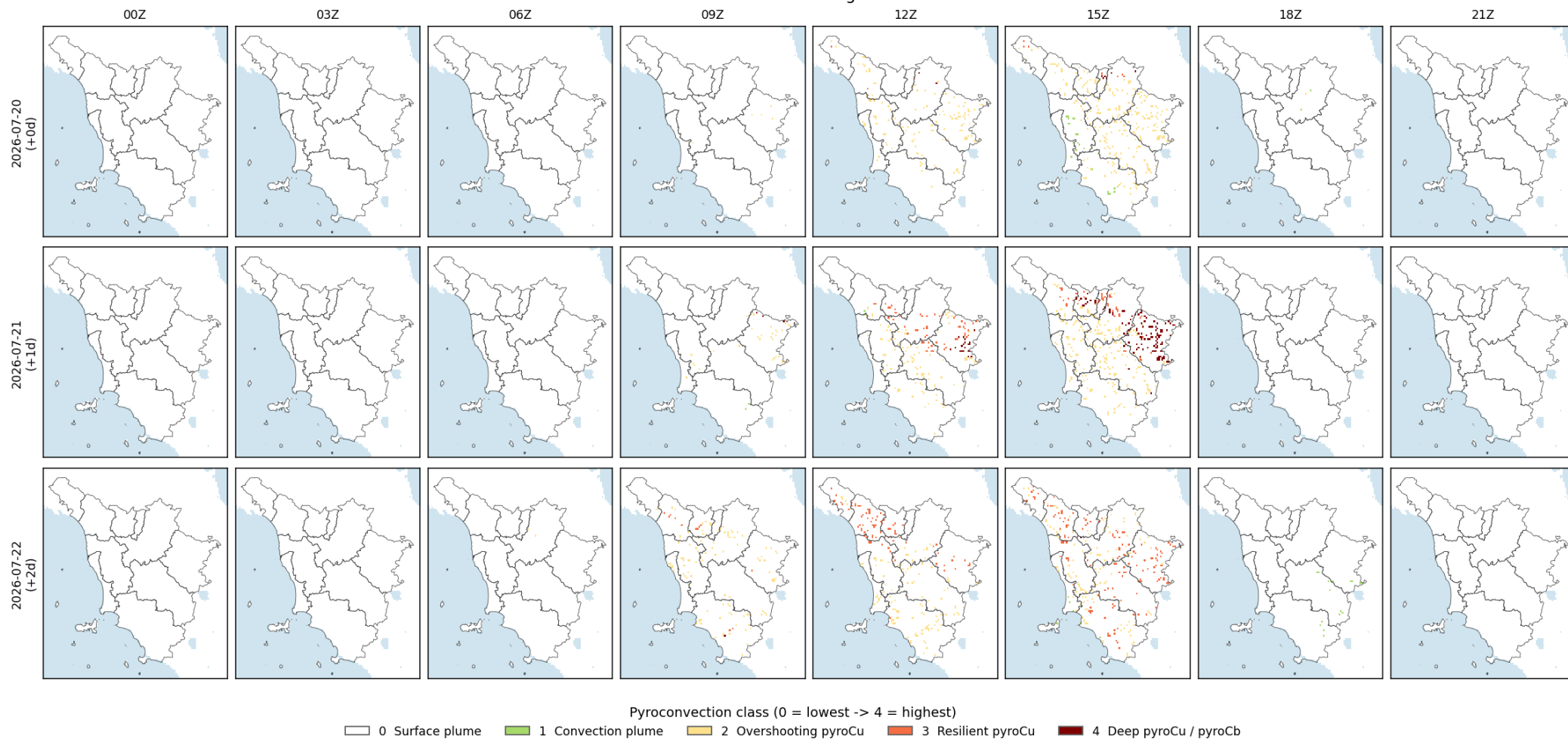


Figure 1: gated

Tuscany pyroconvection -- 3-day forecast -- POTENTIAL -- atmospheric upper bound (assumes a pyroCu-capable fire everywhere)
 ICON-EU model levels + ICON-ZI 2.2 km gate -- run 2026-07-20 00Z

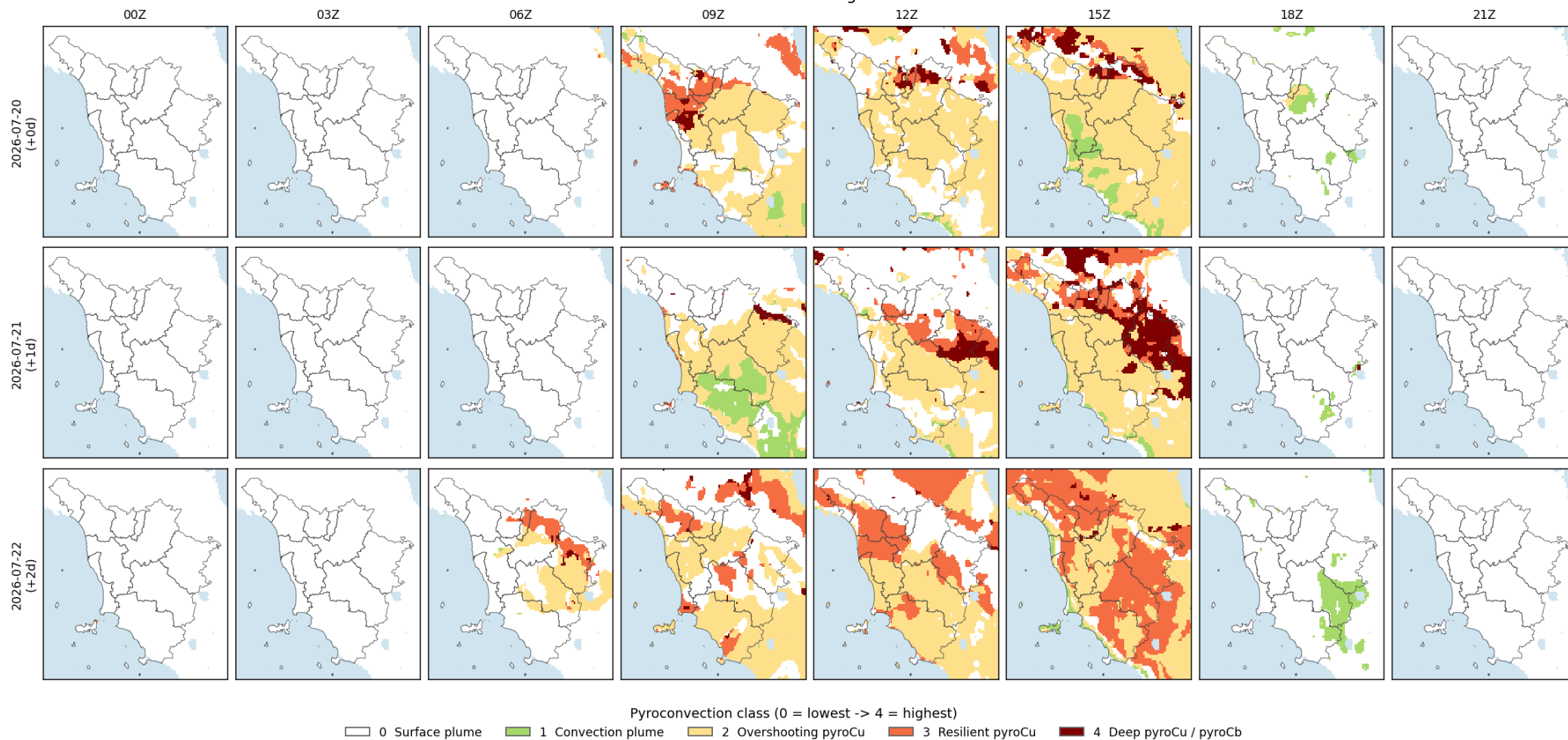


Figure 2: potential